

hw3

Ch 4

1. a) No, 1 generates all of $\mathbb{Z}_{60}$ and 1 is not prime
b) No $U(8)$ is isomorphic to the Klein-4 group which is not cyclic
Additionally, $U(8) = \{1, 3, 5, 7\}$. 1 does not generate, $3^2 \bmod 8 = 1$ so 3 does not generate.
 $5^2 \bmod 8 = 1$ so 5 does not generate. $7^2 \bmod 8 = 1$ so 7 does not generate.
Therefore $U(8)$ is not cyclic
c) No, proof by contradiction. Let $\mathbb{Q} = \langle a \rangle$
Then $\forall x \in \mathbb{Q}, x = na$ where $n \in \mathbb{Z}$. Since $a \in \mathbb{Q}, a = \frac{p}{q}$ where $p, q \in \mathbb{Z}$ and $\gcd(p, q) = 1$. Let $x = P$ where P is a prime bigger than any prime factor of p . Then $P = \frac{np}{q}$, since n is an integer, this would imply that P has factors which it can't since we chose it to be a prime.
Therefore contradiction
d) No look at $U(8)$ from part b. It is not cyclic but every subgroup is cyclic
e) No. Assume group G has finite amount of subgroups. If one of the subgroups is infinite, then it is isomorphic to $\mathbb{Z}$ and therefore has infinite subgroups so if G has finite amount of subgroups, the size of the subgroups must be finite therefore G must be finite as well.
2. a) 12
b) inf
c) inf
d) 4
e) $240/\gcd(72, 240) = 240/24 = 10$
f) $\gcd(312, 471) = \gcd(312, 159) = \gcd(159, 153) = \gcd(6, 153) = \gcd(6, 3) = 3$ so $471/3 = 157$
3. a) all mult of 7 ie $\{7n : n \in \mathbb{Z}\}$
b) $\{15, 6, 21, 3, 18, 9, 12, 0\}$
c)
 $\langle 0 \rangle = \{0\}$
 $\langle 1 \rangle = \{0, \dots, 11\}$
 $\langle 2 \rangle = \{0, 2, 4, 6, 8, 10\}$
 $\langle 3 \rangle = \{0, 3, 6, 9\}$
 $\langle 4 \rangle = \{0, 4, 8\}$
 $\langle 6 \rangle = \{0, 6\}$
d)
 $\langle 0 \rangle = \{0\}$
 $\langle 1 \rangle = \{0, \dots, 59\}$
 $\langle 2 \rangle = \{0, 2, 4, 6, 8 \dots 58\}$
 $\langle 3 \rangle = \{0, 3, 6, 9 \dots 57\}$
 $\langle 4 \rangle = \{0, 4, 8 \dots 56\}$

$$\langle 5 \rangle = \{0, 5 \dots 55\}$$

$$\langle 6 \rangle = \{0, 6 \dots 54\}$$

$$\langle 10 \rangle = \{0, 10 \dots 50\}$$

$$\langle 12 \rangle = \{0, 12 \dots 48\}$$

$$\langle 15 \rangle = \{0, 15 \dots 45\}$$

$$\langle 20 \rangle = \{0, 20, 40\}$$

$$\langle 30 \rangle = \{0, 30\}$$

e) Trivial subgroups, $\{0\}$ and $\mathbb{Z}_{13}$

f)

$$\langle 0 \rangle = \{0\}$$

$$\langle 1 \rangle = \{0, \dots, 47\}$$

$$\langle 2 \rangle = \{0, 2, 4, 6, 8 \dots 46\}$$

$$\langle 3 \rangle = \{0, 3, 6, 9 \dots 45\}$$

$$\langle 4 \rangle = \{0, 4, 8 \dots 44\}$$

$$\langle 6 \rangle = \{0, 6 \dots 42\}$$

$$\langle 8 \rangle = \{0, 8 \dots 40\}$$

$$\langle 12 \rangle = \{0, 12, 24, 36\}$$

$$\langle 15 \rangle = \{0, 16, 32\}$$

$$\langle 24 \rangle = \{0, 24\}$$

g) $\{3, 9, 7, 1\}$

h) $\{5, 7, 17, 13, 11, 1\}$

i) $\{7^n : n \in \mathbb{Z}\}$

j) $\{i, -1, -i, 1\}$

k) $\{(2i)^n : n \in \mathbb{Z}\}$

l) $\{\frac{1+i}{\sqrt{2}}, i, \frac{1+i}{\sqrt{2}}i, -1, -\frac{1+i}{\sqrt{2}}, -i, -\frac{1+i}{\sqrt{2}}i, 1\}$

m) $\{\pm 1, \frac{\pm 1 \pm i\sqrt{3}}{2}\}$ (6 total elems)

4. $\left\{ \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}, \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \right\}$ b) $\left\{ \begin{pmatrix} 0 & 1/3 \\ 3 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \right\}$
c) $\left\{ \begin{pmatrix} 1 & -1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}, \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}, \begin{pmatrix} -1 & 1 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ -1 & 1 \end{pmatrix}, I_2 \right\}$ d) $\left\{ \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix} : n \in \mathbb{Z} \right\}$
e) $\left\{ \begin{pmatrix} F(n+2) & F(n+1) \\ -F(n+1) & F(n) \end{pmatrix} : F \text{ is fibonacci seq + } n \in \{0, 1, \dots\} \right\} \cup I_2$
f) Note $a = \frac{\sqrt{3}}{2}$ + $b = \frac{1}{2}$ $\left\{ \begin{pmatrix} a & b \\ -b & a \end{pmatrix}, \begin{pmatrix} b & a \\ -a & b \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \begin{pmatrix} -a & -b \\ -b & -a \end{pmatrix}, \begin{pmatrix} a & -b \\ b & a \end{pmatrix}, I_2 \right\}$

4.

5. $|\langle 1 \rangle| = |\langle 17 \rangle| = |\langle 5 \rangle| = |\langle 13 \rangle| = |\langle 7 \rangle| = |\langle 11 \rangle| = 18$

$$|\langle 2 \rangle| = |\langle 16 \rangle| = |\langle 4 \rangle| = |\langle 14 \rangle| = |\langle 8 \rangle| = |\langle 10 \rangle| = 9$$

$$|\langle 3 \rangle| = |\langle 15 \rangle| = 6$$

$$|\langle 6 \rangle| = |\langle 12 \rangle| = 3$$

$$|\langle 9 \rangle| = 2$$

$$|\langle 0 \rangle| = 1$$

6. Elements r_{90}, r_{180}, r_{270} which are rotations, D_1, D_2 which are diagonal flips, e which is stay same, V and H which are vertical and horizontal flips

$$|<r_{90}>| = |<r_{270}>| = 4$$

$$|<r_{180}>| = |<D_1>| = |<D_2>| = |<V>| = |<H>| = 2$$

$$|<e>| = 1$$

7. $Q_8 = \{1, -1, i, -i, j, -j, k, -k\}$

$$<1> = \{1\}$$

$$<-1> = \{-1, 1\}$$

$$<i> = \{i, -1, -i, 1\}$$

$$<j> = \{j, -1, -j, 1\}$$

$$<k> = \{k, -1, -k, 1\}$$

8. $U(30) = \{1, 7, 11, 13, 17, 19, 23, 29\}$

$$<1> = \{1\}$$

$$<7> = \{1, 7, 19, 13\}$$

$$<11> = \{1, 11\}$$

$$<13> = \{1, 13, 19, 7\}$$

$$<17> = \{1, 17, 19, 23\}$$

$$<19> = \{1, 19\}$$

$$<23> = \{1, 17, 19, 23\}$$

$$<29> = \{1, 29\}$$

$$<11, 29> = \{1, 11, 19, 29\}$$

Ch 6

1. Because if g has order 5 and h has order 7, By lagrange's theorem the order of a subgroup must divide the order of the group so 5 and 7 divide G and the smallest such number with 5 and 7 as factors is 35 so $|G| \geq 35$
2. 1, 2, 3, 4, 5, 6, 10, 12, 15, 20, 30, 60
3. False, let $G = \mathbb{Z}$ and $H = \{0\}$, the trivial subgroup. H has infinite index as we can take any element $x \in \mathbb{Z}$ and do $x + \{0\} = \{x\}$
4. False, the subgroup $2\mathbb{Z}$ of $\mathbb{Z}$ has infinite order
5. a) $\langle 8 \rangle = \{0, 8, 16\}$, $\langle 8 \rangle + 1 = \{1, 9, 17\}$, $\langle 8 \rangle + 2$, $\langle 8 \rangle + 3$, ... $\langle 8 \rangle + 7$. Left and right cosets the same since addition is commutative
 b) $U(8) = \{1, 3, 5, 7\}$, $\langle 3 \rangle = \{3, 1\}$
 $\langle 3 \rangle$
 $\langle 3 \rangle^5 = \{5, 15\} = \{5, 7\}$
Again, left and right are same since mult is commutative
 c) $3\mathbb{Z}$, $3\mathbb{Z} + 1$, $3\mathbb{Z} + 2$. Left and right are same since addition is commutative
 d) A_4 is all even permutations and S_4 is all permutations.
 So $|S_4| = 24$ and $|A_4| = 24/2 = 12$

SO there must be 2 cosets, the trivial one is itself.

The other coset must be odd permutations. of 4 elements. This will be true for both left and right coset

The cosets can be written as A_4 and $(1, 2)A_4$ for left and A_4 and $A_4(1, 2)$ for right

e) Same logic as before.

For left: A_n and $(1, 2)A_n$, even and odd permutations respectively

For right: A_n and $A_n(1, 2)$, even and odd permutations respectively

f) $D_4 = \{(1), (1234), (13)(24), (1432), (13), (24), (12)(34), (14)(23)\}$

$|D_4| = 8$ so index $= \frac{|S_4|}{|D_4|} = \frac{24}{8} = 3$ so there are 3 cosets. We have one already

Left:

$(12)D_4 = \{(12), (234), (1324), (143), (132), (124), (34), (1423)\}$

$(14)D_4 = \{(14), (123), (1342), (243), (1243), (23), (134), (142)\}$

Right:

$D_4(12) = \{(12), (134), (1423), (243), (34), (1324), (132), (124)\}$

$D_4(14) = \{(14), (123), (1342), (132), (1243), (23), (143), (142)\}$

g) $\mathbb{T}$ is the set of all complex numbers such that their magnitude is 1

The cosets of $\mathbb{T}$ in $\mathbb{C}^*$ would be $\{x \in \mathbb{C}^* : |x| = r\}$ where $r \in \mathbb{R}^+$ so there are uncountably infinite cosets. Also left and right cosets will be the same since abelian

h) There are 8 cosets total (since $|H|=3$)

Left:

$H = \{(1), (123), (132)\}$

$(12)H = \{(12), (13), (23)\}$

$(14)H = \{(14), (1234), (1324)\}$

$(24)H = \{(24), (1423), (1342)\}$

$(34)H = \{(34), (1243), (1432)\}$

$(12)(34)H = \{(12)(34), (143), (243)\}$

$(13)(24)H = \{(13)(24), (142), (234)\}$

$(14)(23)H = \{(14)(23), (124), (134)\}$

Right:

$H = \{(1), (123), (132)\}$

$H(12) = \{(12), (13), (23)\}$

$H(14) = \{(14), (1423), (1432)\}$

$H(24) = \{(24), (1243), (1324)\}$

$H(34) = \{(34), (1234), (1342)\}$

$H(12)(34) = \{(12)(34), (134), (234)\}$

$H(13)(24) = \{(13)(24), (124), (243)\}$

$H(14)(23) = \{(14)(23), (142), (143)\}$

6. The left cosets are all invertible matrices that have that specific determinant. For example let $H = SL_2(\mathbb{R})$ and let $X \in GL_2(\mathbb{R})$ with $\det(X)=x$. Then the left coset $XH = \{N : \det(N)=\det(X)=x\}$.

The index is therefore infinite (uncountably infinite since the determinant can be any number in $\mathbb{R}^*$)

7. $a=4$ and $n=15$, they are coprime. $\phi(15) = 4$ so $4^4 = 216$ and $216 \bmod 15 = 1$ which verifies the statement $4^4 \equiv 1 \pmod{15}$.

8. Theorem: $a^{p-1} \equiv 1 \pmod{p}$ if p prime.

Let the given be true. Let $p=4n+3$ be prime and $\exists$ a solution to $x^2 \equiv -1 \pmod{p}$

Then $a^{p-1} = a^{4n+2} = a^2 a^{4n} \equiv -1(a^{4n}) = -1(a^2)^{2n} \equiv -1(-1)^{2n} = -1$

But then this implies that $a^{p-1} \equiv -1 \pmod{p}$ but the theorem states $a^{p-1} \equiv 1 \pmod{p}$ so this is a contradiction therefore the equation must not have a solution